ABDESSAMAD AIT BRAHIM



AI Engineering Student | Machine Learning & Data Analytics

Seeking a final-year internship (PFE) starting January 2027

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linkedin/abdessamad-aitbrahim **github/abdessamadaitbrahim** **Portfolio**

PROFILE

Final-year engineering student in **Big Data & Artificial Intelligence** at ENSA Tetouan, seeking a **PFE internship starting January 2027**. Hands-on experience in **data analysis, machine learning, NLP, and AI application development**. Skilled in building data pipelines, cleaning and validating datasets, developing predictive models and REST APIs, and designing analytical dashboards. Proficient in **Python, SQL, Pandas, Scikit-learn, PyTorch, and TensorFlow**, with an interest in applying AI and data-driven methods to real-world business challenges.

PROFESSIONAL EXPERIENCE

AI Engineer Intern — LLM Conversational Agent (POC) August 2026

STMicroelectronics · Bouskoura, Casablanca, Morocco

- Developed a conversational AI proof of concept to support the analysis of industrial production data.
- Built Python/Pandas pipelines for **data cleaning, validation**, computation of production capacity and flow indicators.
- Integrated **MariaDB, FastAPI, and Dify** to expose structured data through a read-only REST API.
- Improved response reliability through data quality checks and prompt engineering.

AI Engineer Intern — Labex / Botground June – July 2026

SaaS Project — Sentiment Analysis · Casablanca, Morocco

- Built a **multilingual NLP** pipeline (French, English, Darija) for chatbot conversation classification.
- Annotated a dataset of **4,000+ conversations** across 3 domains: hospitality, logistics, customer support.
- Fine-tuned and benchmarked **XLM-RoBERTa, mBERT, DarijaBERT**, reaching up to **84.3%** accuracy.
- Developed an analytics dashboard (Flask, SQLite, Chart.js) and deployed it on **AWS EC2/S3**.

AI Engineer Intern — SmartMatchRH July – August 2025

Cloud Marketing Hub · Tangier, Morocco (On site)

- Developed a **candidate/job matching** model (Deep Learning & multilingual NLP, Python/TensorFlow).
- Built an end-to-end data pipeline (ingestion, cleaning, feature engineering) served through a **Flask REST API**.
- Designed **3 Power BI dashboards** for HR KPI tracking.
- Results: **+35%** matching accuracy, **+85%** gain in pre-screening, **78%** interview success rate, application volume **×2**.

EDUCATION

Engineering Degree — Big Data & Artificial Intelligence 2024 – Present

National School of Applied Sciences of Tetouan (ENSA Tetouan)

Integrated Preparatory Years for Engineering Schools 2022 – 2024

National School of Applied Sciences of Tetouan (ENSA Tetouan)

Scientific Baccalaureate — French Option 2021 – 2022

Lycée Tarik Ibn Ziad, Kelaat M'Gouna, Tinghir Province

PROJECTS DATA & AI

Deep Learning : Facial Emotion Recognition

- CNNs on 49,779 images: ResNet (**75.4%**) outperforms VGG (66.7%); data augmentation and residual connections (PyTorch/TensorFlow)

Deep Learning : Image Caption Generation

- VGG16/VGG19 encoders + LSTM/GRU/Transformer decoders with attention mechanism (TensorFlow)

Business Intelligence & Data Warehouse

- End-to-end BI solution: SSIS ETL, star schema, SSAS OLAP cube, SQL Server and 3 **Power BI** dashboards

TECHNICAL SKILLS

Programming:	Python, SQL
Data Analysis:	Pandas, NumPy, data cleaning, data validation, feature engineering, exploratory data analysis
Machine Learning:	Scikit-learn, XGBoost, TensorFlow, PyTorch, model training and evaluation
AI & NLP:	Hugging Face Transformers, multilingual NLP, LLM applications, prompt engineering
Databases:	PostgreSQL, MySQL, MariaDB, SQL Server
Backend & BI:	FastAPI, Flask, REST APIs, Power BI, Streamlit
Data Engineering:	ETL/ELT, PySpark, SSIS, SSAS
Tools & Cloud:	Git, Docker, Linux, AWS (EC2, S3)

LANGUAGES

• **Tamazight** (Native) • **Arabic** (Fluent) • **French** (Fluent) • **English** (Fluent) • **Spanish** (Beginner)